Part C. Incident Response Runbooks

Ironbridge National Bank

Use this document after selecting the appropriate playbook. It contains the concrete investigation procedures and commands used to collect evidence and support the escalation/closure decision.

Runbook 1: PHISHING / BUSINESS EMAIL COMPROMISE

Investigation Steps

Step 1 — Examine the Email
Record:

- Sender address
- Sender domain
- Recipient
- Subject
- Date/time received
- URLs
- Attachment names/types
- Urgency or social-engineering language
- Whether the user clicked anything
- Whether credentials were entered

Step 2 — Search Failed Authentication Events
Splunk:

```
index=idx_team4

("failed" OR "failure" OR "4625")

| sort \_time
```

Step 3 — Search Successful Windows Logins

```
index=idx_team4 EventCode=4624

| table \_time host user src_ip Logon_Type

| sort - \_time
```

Step 4 — Investigate the Specific User

```
index=idx_team4 user="\<username>"

| sort \_time

| table \_time host source EventCode user src_ip
```

Decision Handoff

Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 2: MALWARE INFECTION

Linux Investigation

Step 1 — Check Processes
ps aux

Step 2 — Check Network Connections
sudo ss -tulpn

Step 3 — Check Running Services
systemctl --type=service --state=running

Step 4 — Check Scheduled Tasks
crontab -l

sudo ls -la /etc/cron.*

Step 5 — Check Recently Modified Files
sudo find /tmp /var/tmp /home -type f -mtime -1 -ls

Windows Investigation

Step 1 — Processes
Get-Process

Step 2 — Network Connections
Get-NetTCPConnection

Step 3 — Services
Get-Service | Where-Object {$_.Status -eq "Running"}

Step 4 — Scheduled Tasks
Get-ScheduledTask | Where-Object {$_.State -ne "Disabled"}

Step 5 — Defender Detections
Get-MpThreatDetection

Splunk Investigation
index=idx_team4

("malware" OR "threat" OR "defender")

| sort - _time

Decision Handoff
Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 3: RANSOMWARE

Windows Investigation

Step 1 — Find Recently Modified Files
Get-ChildItem \<SHARE_PATH> -Recurse -File |

Sort-Object LastWriteTime -Descending |

Select-Object -First 50 FullName,LastWriteTime

Replace <SHARE_PATH> with the actual bank-share location.

Step 2 — Examine File Extensions
Get-ChildItem \<SHARE_PATH> -Recurse -File |

Group-Object Extension |

Sort-Object Count -Descending |

Select-Object Count,Name

Step 3 — Look for Potential Ransom Notes
Get-ChildItem \<SHARE_PATH> -Recurse -File |

Where-Object {$_.Name -match "README|RECOVER|DECRYPT|RESTORE"}

Splunk Investigation

index=idx_team4

| search ("file" OR "object")

| timechart span=1m count

Decision Handoff

Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 4: DISTRIBUTED DENIAL OF SERVICE (DDoS)

Linux Investigation

Step 1 — View Connections
sudo ss -tn

Step 2 — Count Established Connections
sudo ss -tn state established | wc -l

Step 3 — Check System Resources
top

Step 4 — Check Web Service
Apache:

```
systemctl status apache2
```

Nginx:

```
systemctl status nginx
```

Splunk Investigation

Traffic Over Time

```
index=idx_team4

| timechart span=1m count
```

Top Sources

```
index=idx_team4

| stats count by src_ip

| sort - count
```

HTTP Errors

```
index=idx_team4 status>=400

| timechart span=1m count by status
```

Decision Handoff

Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 5: WEB APPLICATION ATTACK / DEFACEMENT

Linux Investigation

Step 1 — Find Recently Modified Web Files

```
sudo find /var/www -type f -mtime -1 -ls
```

Step 2 — Check Ownership and Permissions

```
ls -la /var/www/html
```

Step 3 — Generate Current File Hashes

```
sudo find /var/www/html -type f -exec sha256sum {} \\;
```

Compare these against your known-good baseline.

Step 4 — Review Access Logs

```
sudo tail -n 100 /var/log/apache2/access.log
```

Step 5 — Review Error Logs

```
sudo tail -n 100 /var/log/apache2/error.log
```

Splunk Investigation

index=idx_team4

(source="*access*" OR sourcetype="*access*")

| stats count by src_ip status

| sort - count

Review errors:

index=idx_team4 status>=400

| table _time src_ip method uri status

| sort - _time

Decision Handoff

Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 6: BRUTE FORCE / CREDENTIAL STUFFING

Windows/Splunk Investigation

Failed Logins

index=idx_team4 EventCode=4625

| stats count earliest(_time) as first latest(_time) as last by user src_ip

| sort - count

Account Lockouts

index=idx_team4 EventCode=4740

| table _time host user

Successful Authentication

index=idx_team4 EventCode=4624

| table _time user src_ip host

| sort - _time

Linux Investigation

Failed SSH

sudo grep "Failed password" /var/log/auth.log

Successful SSH

sudo grep "Accepted" /var/log/auth.log

Count Failures by Source

sudo grep "Failed password" /var/log/auth.log |

```
awk '{print $(NF-3)}' |

sort |

uniq -c |

sort -nr
```

Decision Handoff

Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 7: AI DEEPFAKE-ENABLED FRAUDULENT TRANSACTION

Investigation

Search the associated account:

```
index=idx_team4 user="\<username>"

| table \_time host source EventCode src_ip

| sort - \_time
```

Record:

- Person supposedly making request
- Communication method
- Requested transaction
- Time
- Amount
- Recipient
- Associated account
- Verification attempts

Decision Handoff

Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 8: INSIDER THREAT

Windows Investigation

Administrators

```
Get-LocalGroupMember -Group "Administrators"
```

Recent Security Events
Get-WinEvent -LogName Security -MaxEvents 100

Account Creation
Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4720}

Group Membership Changes
Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4732}

Linux Investigation

Sudo Activity
sudo grep "sudo" /var/log/auth.log

Login History
last

Currently Logged-In Users
who

Sudo Membership
getent group sudo

Splunk Investigation
index=idx_team4

(EventCode=4720 OR EventCode=4732)

| table _time host user EventCode

| sort - _time

Decision Handoff
Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 9: DATA EXFILTRATION

Linux Investigation

All Connections
sudo ss -tunap

Established TCP Connections
sudo ss -tn state established

Processes Using Network Connections
sudo lsof -i

Windows Investigation
Get-NetTCPConnection -State Established |

Select-Object LocalAddress,LocalPort,RemoteAddress,RemotePort,OwningProcess

If a suspicious PID is identified:

Get-Process -Id \<PID>

Splunk Investigation

Top Destinations
index=idx_team4

| stats count by dest_ip

| sort - count

Outbound Data Volume
If byte-count fields are available:

index=idx_team4

| stats sum(bytes_out) as bytes_out by src_ip dest_ip

| sort - bytes_out

Decision Handoff
Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Runbook 10: PRIVILEGE ESCALATION / LATERAL MOVEMENT

Linux Investigation

Check Sudo Users
getent group sudo

Authentication and Privilege Activity
sudo grep -E "sudo|Accepted|Failed" /var/log/auth.log

Login History
last

Processes
ps aux

Connections
sudo ss -tunap

Compare all results against your known-good A2/A3 baseline.

Windows Investigation

Administrators
Get-LocalGroupMember -Group "Administrators"

New Accounts
Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4720}

Group Changes
Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4732}

Processes
Get-Process

Services
Get-Service

Scheduled Tasks
Get-ScheduledTask

Network Connections
Get-NetTCPConnection

Splunk Investigation

New Accounts / Group Changes
index=idx_team4

(EventCode=4720 OR EventCode=4732)

| table _time host EventCode user

| sort - _time

Authentication
index=idx_team4

(EventCode=4624 OR EventCode=4625)

| table _time host EventCode user src_ip

| sort - _time

Pay particular attention to unexpected activity involving:

10.10.60.197 → 10.10.60.196

Decision Handoff

Review the collected evidence against the companion playbook's Incident Decision Criteria and Decision Point. Escalate or close according to that playbook; do not make the decision from a single indicator alone.

Standard Incident Documentation

For every investigation, record:

- Incident Name
- Date
- Start Time
- Severity
- Analyst
- Affected Host(s)
- Affected Account(s)
- Detection Source
- Initial Alert
- Playbook Used
- Current Status

Major Investigation Step Record

- Timestamp
- Analyst
- Host
- Command/Splunk Query
- Result
- Decision
- Action Taken
- Evidence Preserved